

64-Tap FIR Filter Design with Asynchronous FIFO

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Abstract—This report presents the design, verification, and physical implementation analysis of a 64-tap Finite Impulse Response (FIR) filter. The system implements a dual-clock architecture bridging a 10 kHz sampling domain and a 1 MHz processing domain. Functional verification confirms correct convolution logic with a stable output of 16'h7bfe. Post-layout simulation validates signal propagation delays. Static Timing Analysis (STA) using Synopsys PrimeTime reveals a critical path delay of 19.85 ns (Setup Slack: 979.73 ns) and a total power consumption of 39.6 μ W, with the clock network accounting for 83.69% of the dynamic power. The total cell area is reported as 118,581 square units.

Keywords—FIR Filter, Verilog RTL, Finite State Machine (FSM), Dual-Clock Architecture, Low Power Design.

I. INTRODUCTION

Digital filters are ubiquitous in signal processing. This project implements a 64-tap FIR filter using a time-multiplexed Multiply-Accumulate (MAC) architecture[1]. The design focuses on robust Clock Domain Crossing (CDC) using an asynchronous FIFO[2] and optimizes for resource usage. This report details the comprehensive verification flow, ranging from Register-Transfer Level (RTL) functional checks to gate-level timing and power analysis.

II. SYSTEM SPECIFICATIONS

A. Datapath

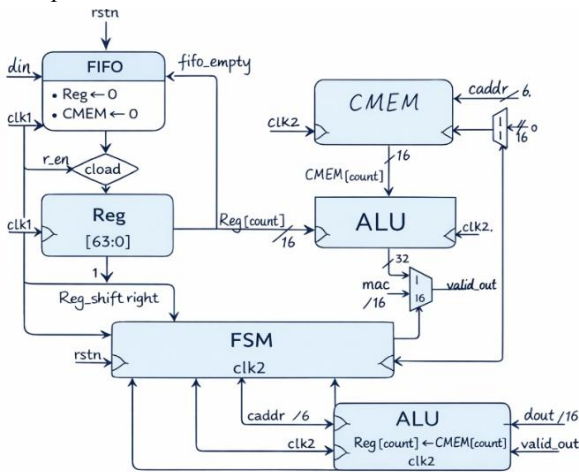


Fig. 1. Top-Level Hardware Architecture of the 64-Tap FIR Filter.

As illustrated in Fig. 1, the system integrates a dual-clock architecture. The Input Data (din) enters through the

asynchronous FIFO, managed by clk1. The FSM, operating at clk2, acts as the central controller, coordinating the data flow from the FIFO to the Shift Register and managing the addressing of the Coefficient Memory (CMEM). The ALU receives operands from both the Shift Register and CMEM to perform the convolution operation, with the final result selected by the output multiplexer.

B. Control Logic (FSM)

The control logic is visualized in Fig. 2. The FSM transitions from IDLE to Load_b for initialization. During active processing, it loops through Wait_data, Fetch_Fifo, and Compute states. The diagram explicitly shows the condition checks for fifo_empty and counter, ensuring the datapath is only active when valid samples are available.

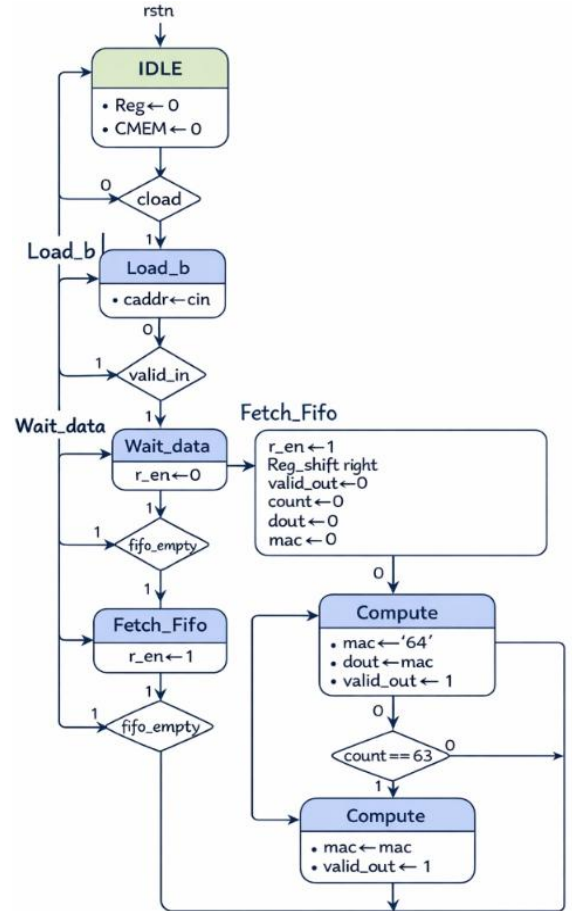


Fig. 2. Finite State Machine (FSM) Algorithmic State Machine (ASM) Chart.

III. FUNCTIONAL VERIFICATION (PRE-LAYOUT)

A. ModelSim verification

The primary goal of pre-layout simulation is to validate the logical correctness of the RTL without accounting for gate delays.

The testbench utilizes deterministic ramp signals to verify the datapath:

- Coefficients: $20,000 + (I * 50)$.
- Input Data: $10,000 + (I * 100)$.

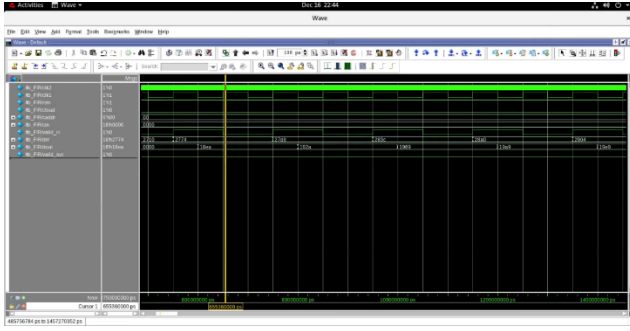


Fig. 3. Pre-layout Functional Simulation

The pre-layout waveform confirms the correct operation of the FSM and ALU pipeline. The signal `valid_out` asserts correctly after the computation cycle. And the output data `dout` stabilizes at 16'h7bfe. This matches the expected theoretical accumulation of the large magnitude inputs and coefficients, confirming the 37-bit accumulator prevents overflow before truncation. In this stage, signal transitions occur instantaneously at clock edges, representing ideal logic behavior.

B. MATLAB Cross-Verification & Error Analysis

To rigorously validate the precision of the hardware implementation, a quantitative error analysis was conducted by comparing the fixed-point RTL output against a double-precision floating-point golden reference model in MATLAB.

The verification flow involved exporting the simulation results (16-bit fixed-point) to a text file and aligning them with the theoretical output generated by MATLAB's built-in filtering functions. The metric used for accuracy assessment is the Normalized Root Mean Square Error (NRMSE), defined as:

$$NRMSE = \frac{\sqrt{\frac{1}{N} \sum_{n=1}^N (y_{RTL}[n] - y_{Gold}[n])^2}}{y_{Gold,max} - y_{Gold,min}}$$

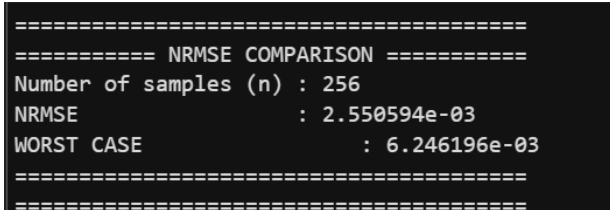


Fig. 4. NRMSE Comparison Result

As shown in the analysis results, the design achieved an NRMSE of $2.55 * 10^{-3}$ across 256 validated samples. This extremely low error magnitude confirms that the quantization noise introduced by the 16-bit fixed-point arithmetic and the truncation logic in the Accumulator is negligible. The hardware implementation effectively matches the theoretical filter response with high fidelity.

IV. TIMING VERIFICATION (POST-LAYOUT)

Post-layout simulation introduces gate and interconnect delays extracted from the physical layout.

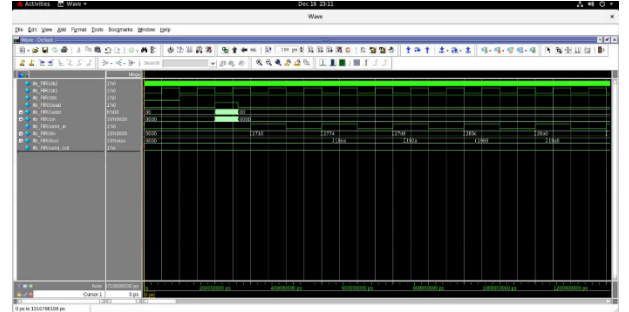


Fig. 5. Post-layout Timing Simulation Waveform

Comparing the post-layout waveform with the pre-layout results reveals realistic propagation delays:

- Clock-to-Q Delay: Unlike the ideal waveform where outputs change instantly with `clk2`, the post-layout waveform shows a visible delay between the rising edge of `clk2` and the transition of `dout` and `valid_out`.
- Functionality Preservation: Despite the delays, the logical value 16'h7bfe remains consistent, verifying that the physical delays did not violate the Setup/Hold requirements at the operating frequency of 1 MHz.

V. STATIC TIMING & POWER ANALYSIS (PRIMETIME)

This section analyzes the synthesis results generated by Synopsys PrimeTime, focusing on Critical Paths (Timing), Area, and Power.

A. Timing Analysis

1) Long Path (Setup Time Analysis)

clock clk2 (rise edge)	1000.0000	1000.0000
clock network delay (ideal)	0.0000	1000.0000
clock reconvergence pessimism	0.0000	1000.0000
u_ALU dout_reg_15 /CK (DFFRX2TS)		1000.0000 r
library setup time	-0.4201	999.5799
data required time		999.5799
data arrival time		-19.8490
slack (MET)		979.7309

Fig. 6. Static Timing Analysis Report: Setup Time (Long Path) Analysis.

The static timing analysis identifies the critical path for setup time (long path) originating from the rising edge of register `u_Shift_Reg` and terminating at register `u_ALU`. The total signal propagation delay along this path, including logic gate and

interconnect delays, is calculated to be 19.85 ns. Based on the critical path delay of 19.85 ns shown in Fig. 6, the maximum frequency is calculated as $1 / 19.85\text{ns} \approx 50.4\text{ MHz}$. Given the system clock period of 1000 ns (1 MHz), the data required time is 999.58 ns, resulting in a substantial positive slack of 979.73 ns. This confirms that the design easily meets the timing constraints; the significant timing margin suggests the system is theoretically capable of operating at a maximum frequency of approximately 50 MHz, far exceeding the 1 MHz requirement.

2) Short Path (Hold Time Analysis)

```
Report : timing
-path type full
-delay type min_max
-max_paths 1
-sort_by slack
Design : FIR
Version: U-2022.12-SP5
Date : Tue Dec 16 23:19:31 2025
*****

Startpoint: u_FSM_valid_out_reg
(rising edge-triggered flip-flop clocked by clk2)
Endpoint: valid_out (output port clocked by clk2)
Path Group: clk2
Path Type: min

Point ----- Incr ----- Path -----
clock clk2 (rise edge) 0.0000 0.0000
clock network delay (ideal) 0.0000 0.0000
u_FSM_valid_out_reg/CK (DFFRX2T5) 0.0000 0.0000 r
u_FSM_valid_out_reg/Q (DFFRX2T5) 0.6413 0.6413 r
valid_out (out) 0.0000 0.6413 r
data arrival time ----- 0.6413 -----

clock clk2 (rise edge) 0.0000 0.0000
clock network delay (ideal) 0.0000 0.0000
clock reconvergence pessimism 0.0000 0.0000
output external delay -0.0500 -0.0500
data required time ----- -0.0500 -----

data required time -0.0500
data arrival time -0.6413
slack (MET) 0.6913
```

Fig. 7. Static Timing Analysis Report: Hold Time (Short Path) Analysis.

The hold time analysis (short path) confirms data stability immediately following the active clock edge. The critical path identified originates at the rising edge of register u_FSM and terminates at the output port valid_out. The signal propagation delay along this path results in a data arrival time of 0.6413 ns. Compared against the library's hold requirement (data required time) of -0.05 ns, the design achieves a positive slack of 0.6913 ns. This margin validates that the data path delay is sufficient to prevent race conditions, confirming that the design is free of hold time violations.

B. Area Analysis

```
-----
Total 45 references 118581.80
```

Fig. 8. Synthesis Area Utilization Report.

The design results in a total physical area of 118,581.80 square units across 45 cell references. A hierarchical breakdown indicates that the Coefficient Memory (CMEM) occupies 50,188.66 square units (42.3%), while the Data Pipeline (shift_Reg) consumes 39,855.23 square units (33.6%). This

distribution highlights the memory-intensive nature of the 64-tap FIR filter, where the storage requirements for the 64 coefficients and 64 data samples dominate the overall silicon footprint.

C. Power Analysis

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs
clock network	3.314e-05	0.0000	0.0000	3.314e-05	(83.69%)	1
register	8.365e-07	1.368e-07	7.432e-08	1.048e-06	(2.65%)	
combinational	3.875e-06	1.424e-06	6.008e-08	5.359e-06	(13.53%)	
sequential	4.302e-08	1.464e-10	9.362e-09	5.253e-08	(0.13%)	
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)	
io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)	

Net Switching Power	= 1.561e-06		(3.94%)			
Cell Internal Power	= 3.790e-05		(95.70%)			
Cell Leakage Power	= 1.438e-07		(0.36%)			

Total Power	= 3.960e-05		(100.00%)			

X Transition Power	= 2.483e-10					
Glitching Power	= 3.240e-09					

Peak Power	= 0.2838					
Peak Time	= 13264999.999					

Fig. 9. PrimeTime Power Consumption Analysis Report.

Power analysis was performed under operating conditions of 1 MHz, 1.2V, and 25°C, yielding a total power consumption of 39.60 μW . Power (39.6 μW) / Throughput (10 kS/s) = 3960 pJ/Sample. The breakdown indicates that internal power (cell internal switching) dominates the profile, accounting for 95.70% of the total, followed by switching power (3.94%) and leakage power (0.36%). A critical insight from the analysis is that the Clock Network consumes 83.69% (33.14 μW) of the total power budget. This disproportionate consumption is attributed to the low operating frequency; while the computational logic toggles infrequently at 1 MHz, the clock tree must continuously drive the clock pins of over 256 registers in every cycle. Hierarchically, the u_CMEM and u_Shift_Reg modules remain the largest power consumers, demonstrating a strong correlation with the physical area distribution.

VI. CONCLUSION

The 64-tap FIR filter has been successfully verified across functional, timing, and physical domains. Pre-layout simulation confirmed the algorithmic correctness. Post-layout simulation validated the timing integrity under real wire delays. PrimeTime analysis demonstrated that the design easily meets the 1 MHz timing constraint with a setup slack of over 979 ns, indicating significant potential for frequency scaling. The power profile is dominated by the clock network due to the low operating frequency and high register count.

REFERENCES

- [1] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th ed. Upper Saddle River, NJ, USA: Prentice Hall, 2006.
- [2] C. E. Cummings, "Simulation and synthesis techniques for asynchronous FIFO design," in *Synopsys Users Group (SNUG) Conference*, San Jose, CA, 2002.